

➤ CircuitGuard-PCB Defect Detection and Classification Application

Internship Project Report

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“A Deep Learning Based PCB Defect Detection System with Streamlit Web Integration”

Abstract

This project, **CircuitGuard**, focuses on developing an intelligent system for automated detection and classification of defects in Printed Circuit Boards (PCBs). By leveraging image processing and deep learning techniques, the model identifies defects such as missing holes, open circuits, short spurs, spurious copper, and mouse bites. The project integrates both backend processing and a Streamlit-based frontend for interactive visualization of detected defects. The end goal is to assist PCB manufacturers in automating quality assurance and reducing manual inspection time.

1. Introduction

Printed Circuit Boards (PCBs) are the foundation of all modern electronic devices. Ensuring their quality is critical, as even a minor defect can lead to failure in entire systems. Traditional manual inspection is time-consuming and error-prone. The CircuitGuard project was developed to automate the defect detection process using machine learning and image processing. The system uses reference-based image subtraction to highlight defects and a Convolutional Neural Network (CNN) model to classify them accurately. The final deployment includes a user-friendly Streamlit web interface for uploading PCB images and viewing defect annotations in real time.

2. Dataset Creation and Methodology

The dataset was created manually using image subtraction between defect-free ('good') and defective PCB images. The subtraction highlights defect regions clearly, which were then used for classification. The dataset includes five defect categories — Missing Hole, Mouse Bite, Open Circuit, Short Spur, and Spurious Copper.

Dataset Structure:

1. Good PCB Images – defect-free reference boards.
2. Defective PCB Images – boards containing one or more defects.
3. Subtracted Output – defect-highlighted maps created using OpenCV subtraction and thresholding.

The methodology involves the following stages:

- Image Subtraction and Thresholding using OpenCV.
- Contour Extraction and ROI Segmentation.
- Model Training with VGG16 (transfer learning).
- Testing and Evaluation using unseen images.
- Frontend & Backend integration using Streamlit.

3. Implementation Details

The backend system was implemented in Python using OpenCV, TensorFlow, and Keras. The model used is VGG16 pre-trained on ImageNet, fine-tuned on the PCB defect dataset. Image generators were used to preprocess data, augment training samples, and feed batches efficiently during model training. The Streamlit frontend provides an interactive UI to upload PCB images, view defect highlights, and download annotated results.

Training Process:

- Loaded ROI-labeled dataset from CSV.
- Split data into 80/20 train-test ratio.
- Trained VGG16 model with a custom dense layer head.
- Fine-tuned top layers for improved classification accuracy.
- Evaluated model using confusion matrix and accuracy metrics.

Testing and Evaluation:

- Test data loaded and evaluated using ImageDataGenerator.
- Model achieved over 97% accuracy on unseen test data.
- Evaluated performance using loss and accuracy metrics.

4. Frontend and Backend Integration

The frontend of CircuitGuard was developed using **Streamlit**. It allows users to upload both the original and defective PCB images, visualize detection progress, and see summarized defect reports. The backend integrates OpenCV for image subtraction and TensorFlow for defect classification. The two components work together to produce annotated output images in real-time.

UI Features:

- Dual image upload interface (original vs defective PCB).
- Animated progress bar and status display.
- Result visualization with highlighted defects.
- Defect summary table (type, count, severity).
- Option to download processed results.

5. Results and Conclusion

The CircuitGuard model successfully detected and classified PCB defects with high accuracy and reliability. The integration of VGG16 and OpenCV proved effective in differentiating minute variations between images. The Streamlit web interface enhanced user accessibility and provided an interactive platform for visual inspection. The system achieved over 97% test accuracy, demonstrating its potential for real-world industrial deployment.

Future Enhancements:

- Incorporate real-time camera-based PCB inspection.
- Expand dataset with multi-angle and multi-light samples.
- Integrate with cloud-based dashboards for defect analytics.
- Optimize model using EfficientNet or lightweight CNNs for faster inference.

Conclusion: The project successfully achieved its objective of creating an intelligent and user-friendly PCB defect detection system. It demonstrates a strong blend of deep learning, image processing, and frontend integration — an example of applied AI in manufacturing.